

Key Terms Glossary

MEDICARE BILLING & ANALYTICS FRAMEWORK REFERENCE

Documentation Companion: Healthcare and Medicare billing involve specialized terminology. This glossary establishes foundational concepts matching strict CMS guidelines.

NPI (National Provider Identifier)

A unique 10-digit number assigned by CMS to every healthcare provider in the United States. The primary identifier for providers in this dataset. Never changes even if the provider moves states or changes specialty.

HCPCS code (Healthcare Common Procedure Coding System)

An alphanumeric code that identifies a specific medical procedure or service for billing purposes. Every service billed to Medicare must have a HCPCS code. Level I codes are CPT codes maintained by the AMA; Level II are created by CMS to track products, supplies, and alternative services.

Billed charge (avg_billed_charge)

The price a provider submits on their claim - set by their internal chargemaster price list. This is not what Medicare pays. It is the starting point for all billing, set high to preserve negotiating leverage with private insurers. Medicare ignores it entirely when calculating payment.

Allowed amount (avg_allowed_amount)

What Medicare's fee schedule says a procedure is worth. Calculated independently of the billed charge using the Resource-Based Relative Value Scale (RBRVS). This is the actual economic transaction amount. The provider cannot collect more than this from Medicare patients regardless of what they billed.

Medicare payment (avg_medicare_payment)

What Medicare actually transfers to the provider - approximately 80% of the allowed amount. The remaining 20% is the patient's copay obligation, which may be covered by Medigap or supplemental insurance. This column shows only Medicare's portion.

Standardized payment (avg_standardized_payment)

The Medicare payment with the Geographic Practice Cost Index (GPCI) adjustment removed. Standardization explicitly eliminates geographic differences in payment rates for individual services, such as those that account for local wages or input prices. This makes Medicare payments across regions comparable so that remaining differences reflect variation in physician practice patterns or beneficiary acuity.

Key Terms Glossary

MEDICARE BILLING & ANALYTICS FRAMEWORK REFERENCE

Billed-to-allowed ratio (Analytical Metric)

The core analytical metric in this series. Calculated as $\text{avg_billed_charge} / \text{avg_allowed_amount}$. A ratio of 3.0x means a provider billed three times what Medicare determined the procedure is worth. Higher ratios indicate greater chargemaster inflation above the fee schedule.

GPCI (Geographic Practice Cost Index)

CMS's adjustment factor for geographic cost differences. Providers in high cost-of-living areas (New York, San Francisco) receive a higher GPCI premium. Rural and low-cost areas receive less. Removing the GPCI via standardized payment reveals whether payment differences between areas are due to cost of living or to other structural factors.

RUCA code (Rural-Urban Commuting Area)

A classification system from the USDA that categorizes zip codes on a scale from 1 (urban core) to 10 (most rural) based on population density and commuting patterns. Used in this project to classify providers as Urban (1-3) or Rural (4-10, including decimal sub-codes 10.1-10.3), following the Health Resources & Services Administration (HRSA) classification standard.

Chargemaster (Internal Provider Price List)

A hospital or provider's internal master list of prices for every service they offer. Prices are set high - often 2-10x the Medicare allowed amount - to preserve negotiating leverage with private insurers. The chargemaster is largely disconnected from what any payer actually pays.

Medicare Part B (Outpatient Physician & Diagnostic Services)

The outpatient physician and diagnostic services component of Medicare. Covers doctor visits, outpatient procedures, laboratory tests, and durable medical equipment. This dataset covers Part B only. Part A (hospital inpatient) and Part D (prescription drugs dispensed at pharmacy) are separate programs not covered here.

CMS suppression (Privacy Data Rule)

To protect the privacy of Medicare beneficiaries, any aggregated records which are derived from 10 or fewer beneficiaries are strictly excluded/suppressed from the dataset. Small rural providers and rare procedures are systematically absent from the data. Every geographic finding in this series should be interpreted with this baseline caveat.

Key Terms Glossary

MEDICARE BILLING & ANALYTICS FRAMEWORK REFERENCE

Analytical grain (Dataset Dimensionality)

The level of detail at which one row in the dataset is unique. In this dataset, the grain is provider x procedure code x service location type. The same provider can legitimately bill the same procedure code in both a facility (F) and office (O) setting - these are two valid separate rows.

Volume-weighted average (Statistical Methodology)

An average that weights each provider's contribution by their service volume, not equally. A provider with 50,000 services contributes proportionally more to the average than one with 11 services. All ratio calculations in this series use volume-weighted averages to prevent low-volume edge cases from distorting findings.
